Curriculum Vitae

SeokHyeon (Lucas) Kong

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RESEARCH INTERESTS

Quantum Computer Architecture, Quantum Error Correction and Detection, Quantum Compiler, Quantum Controller, Quantum Programming Language, Quantum Algorithm Optimization

EDUCATION

Sungkyunkwan University (SKKU) — **Suwon, South Korea**
Bachelor of Science in Electronic and Electrical Engineering — *Expected February 2026*
Bachelor of Science in Advanced Semiconductor Engineering — System Architecture Track
Cumulative GPA: 3.95 / 4.0 | Major GPA: 4.0 / 4.0

The Pennsylvania State University — **University Park, PA**
Exchange Student in Computer Engineering — *Fall 2024*
Grade: 4.0 / 4.0

PUBLICATIONS & PRESENTATION

S. H. Kong, D. Kim, K. Maeng*, and E. Seo*, "Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy," *IEEE Computer Architecture Letters*, under review, 2025

S. H. Kong, C. Park, J. Moon, and D. Dahl*, "Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices," *Poster presented at the Quantum Information Research Support Center (Qcenter)*, Suwon, South Korea, Sep. 2025

S. H. Kong and H. Oh*, "Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system," *Journal of the Korean Institute of Information and Communication Engineering*, vol. 27, no. 10, pp. 1208−1213, Oct. 2023

RESEARCH EXPERIENCE

IonQ — **Remote**
Research Mentee, Advised by Dr. Denny Dahl — *08/2025−09/2025*

- Addressed the Instant Insanity problem on NISQ devices by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods.
- Applied a Variational Quantum Algorithm to the Four Corners Map Coloring problem.

The Pennsylvania State University — **University Park, PA**
Undergraduate Researcher, Advised by Prof. Kiwan Maeng — *08/2024−09/2025*

- Conducted in-depth GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differentially private training.
- Proposed an optimization that resolved the dominant Geometric operation overhead, reducing it from 50% to 19% while preserving Gaussian noise quality.

Arch Lab at SKKU **Suwon, South Korea**
Undergraduate Research Intern, Advised by Prof. Dongmoon Min *07/2025–08/2025*

- Built foundational expertise in Quantum / Cryogenic Computer Systems through literature reviews, technical presentations and academic defenses.
- Investigated quantum computer systems, including superconducting-qubit design, error mechanisms, control and readout techniques as well as cryogenic computer architectures.

POSTECH QCQN Lab **Pohang, South Korea**
Undergraduate Research Intern *01/2025*

- Performed 397 nm fluorescence (Doppler) and 729 nm quadrupole spectroscopy on Ca^+ ions and enhanced system stability through targeted adjustments.
- Implemented micromotion optimization via 2D scanning with DC compensation rods.

REFERENCES

Kiwan Maeng, Ph.D. (Advisor)
Assistant Professor, Computer Science and Engineering, The Pennsylvania State University
Email: kvm6242@psu.edu

Denny Dahl, Ph.D. (Advisor)
Customer Success Director, IonQ
Email: denny.dahl@ionq.co

HONORS & AWARDS

NSF Graduate Research Fellowship Program (GRFP)	*Applied*
Qiskit Global Summer School 2025 – Quantum Excellence (IBM Quantum)	*2025*
Quantum Hackathon – Director's Award	*2025*
President's List (Honor Society at SKKU, **2 consecutive years**)	*2024, 2025*
Dean's List (**5 consecutive semesters**)	*SP2023, FA2023, SP2024, FA2024, SP2025*
Korea-U.S. Student Exchange Scholarship in the field of high-tech industry ($9,000)	*FA2024*

TEACHING EXPERIENCE

Academic Tutor, Sungkyunkwan University (SKKU) **Suwon, South Korea**
Intro to Electromagnetism *Fall 2023*

Teaching Assistant, Sungkyunkwan University (SKKU) **Seoul, South Korea**
Engineering Mathematics 2 *Summer 2023*

TECHNICAL SKILLS

Software: Qiskit, CUDA, C, Python, MATLAB, LaTeX
Hardware: System Verilog, Verilog, Verdi, Virtuoso, Nsight Systems, Nsight Compute